

NANOPARTICLES & NANOSCIENCE

KEY VOCABULARY

Nanoparticle: a particle 1–100 nm across (a few hundred atoms).

Coarse particle: dust, 1×10^{-5} – 2.5×10^{-6} m across.

Surface area : volume: ratio that increases as particles get smaller.

Nanoscience: the study of very small structures and their uses.

COMMON MISCONCEPTIONS

~~Smaller particles have a smaller surface area : volume ratio.~~

✓ Smaller particles have a much LARGER ratio.

~~Nanoparticles behave the same as bulk material.~~

✓ They can have very different properties.

~~Nanoparticles are completely safe.~~

✓ Their effects on health are not yet fully known.

SURFACE AREA : VOLUME

A cube of side 2: $SA = 6 \times (2^2) = 24$, $V = 2^3 = 8$.

Ratio = 24 : 8 = 3 : 1

A cube of side 1: $SA = 6$, $V = 1$.

Ratio = 6 : 1 — much larger!

Smaller cube → bigger surface area : volume ratio.

1

DEFINE

State the size range of a **nanoparticle**. [1 mark]

2

SA : VOLUME

A cube has side 3 nm. **Calculate** its surface area : volume ratio. [2 marks]

Answer: _____ (ratio)

3

WHY USEFUL

Explain why a high surface area : volume ratio makes nanoparticles good catalysts. [2 marks]

4

USES

Give one use of nanoparticles (e.g. in sun cream or medicine). [1 mark]

5

FILL THE GAPS

Complete the sentence using the word bank. [2 marks]

As particles get _____, the surface area : volume ratio gets _____.

larger

smaller

less

more

6

RISKS

Suggest why some people are concerned about using nanoparticles. [2 marks]